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mstsa: A Python Library for Information-Theoretic and Statistical Analysis of EEG Microstate Sequences

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Summary

EEG microstate analysis decomposes a multichannel EEG recording into a discrete symbolic time series: each sample is assigned to one of a small number of voltage topography templates, the microstate maps, typically at sampling rates of 100–500 Hz (Lehmann et al., 1998; Michel & Koenig, 2018; Pascual-Marqui et al., 1995). Back-fitting the templates into the original EEG dataset yields a symbolic time series, the microstate sequence. Many research questions address the temporal structure of these sequences, such as: time series memory (Markovianity), autocorrelations including persistence and long-range correlations, measures of syntax and complexity, and comparison against certain null models.

`mstsa` (Microstate Time Series Analysis) is a Python library that addresses these questions. It provides a comprehensive set of information-theoretic, Markov-chain, and statistical tools for the analysis of discrete symbolic sequences, with EEG microstate data as the primary application. The library exposes six modules:

- **`mstsa`** (core): the classical microstate parameters duration, occurrence, and coverage; basic functions to compute Shannon, Rényi, and Tsallis entropies; extensions such as block, topological (Adler et al., 1965), and dispersion (Rostaghi & Azami, 2016) entropies; entropy rate and excess entropy (Crutchfield & Feldman, 2003); auto-information function (AIF) and partial auto-information function (PAIF) (von Wegner et al., 2017; von Wegner, 2018); active information storage (AIS) (Lizier, 2014); sample entropy for discrete sequences, using exact-match comparison and a runs-tracking algorithm that computes estimates for all template lengths in a single $O(N^2)$ pass (Lake et al., 2002; Richman & Moorman, 2000); Lempel–Ziv complexity (Kaspar & Schuster, 1987; Lempel & Ziv, 1976); detrended fluctuation analysis (DFA) (Peng et al., 1994), rescaled-range (R/S) analysis (Hurst, 1951), and diffusion entropy analysis (DEA) (Scafetta & Grigolini, 2002), applicable to both binary-partition random walks and to indicator functions (one-hot encoding) of individual microstate classes; power spectral densities of indicator functions (Hermann & others, 2024); and random-walk construction from binary partitions (Van De Ville et al., 2010; von Wegner et al., 2016).
- **`markov`**: Conditional and joint transition probability matrices, stationary distribution, continuous-time generator matrix, AIF and sample entropy under first-order Markov models (von Wegner, 2026), surrogate sequence generation (von Wegner et al., 2025), nearest-reversible Markov chain projection (Nielsen & Weber, 2015), and detailed-balance statistics (Lynn et al., 2021; Roldán & Parrondo, 2010).
- **`stats`**: Likelihood-ratio (G) tests for zero-, first-, and second-order Markovianity; j- and jk-homogeneity (stationarity); conditional homogeneity; transition-matrix symmetry (Kullback et al., 1962); geometric sojourn-time distributions; sojourn-time distribution fitting that contrasts power-law, exponential, stretched-exponential, and lognormal candidates via the `powerlaw` package (Alstott et al., 2014; Clauset et al., 2009); and a permutation-based multi-subject transition syntax test (Lehmann et al., 2005).
- **`acd`**: Autoregressive Conditional Duration (ACD) model (Engle & Russell, 1998) of

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sojourn-time sequences, with exponential, Weibull, or lognormal innovations, AIC/BIC-based order selection, a Ljung–Box residual autocorrelation test, ACD surrogate-sequence simulation, and ACF/PACF diagnostic plots — quantifying serial clustering between consecutive sojourn durations, complementary to the marginal-distribution fitting in `stats`.

- **`eeg`**: Wackermann/Palus multichannel EEG complexity measures — effective dimensionality Ω, effective voltage Σ, generalized frequency Φ, and Palus complexity coefficient (Paluš et al., 1992; Wackermann, 1996); and continuous sample entropy (`sample_entropy_cont`), using finite neighbourhood matching (Lake et al., 2002; Richman & Moorman, 2000).
- **`microsynt`**: Python implementation of the Microsynt syntactic analysis method (Artoni et al., 2023).

Computationally demanding routines are accelerated by Numba JIT compilation with persistent disk caching (Lam et al., 2015) and optional CFFI-linked C extensions for block entropy, Lempel–Ziv complexity, and continuous and discrete sample entropy.

Statement of Need

EEG microstate research spans five decades, has received growing attention in recent years, and has produced a rich body of results linking microstate dynamics to cognition, sleep, anaesthesia, and neurological and psychiatric disorders (Michel & Koenig, 2018). The standard analysis pipeline has two stages: (1) identifying the template (or microstate) maps by clustering, followed by back-fitting into the original EEG dataset, (2) analyzing the resulting symbolic sequence.

Stage 1 is well covered by existing software. CARTOOL (Brunet et al., 2011), the EEGLAB microstate toolbox (Poulsen et al., 2018), MICROSTATELAB (Nagabhushan Kalburgi et al., 2023), and the Python library Pycrostates (Férat et al., 2022) all provide clustering and segmentation tools. Pycrostates in particular integrates cleanly with the MNE-Python ecosystem and supports both subject- and group-level analyses.

Stage 2, time series analysis, is only partially covered in current software packages. The conventional summary statistics mean duration, occurrence rate, coverage, and transition probabilities, are widely available in the software packages listed above, but numerous recent developments going beyond marginal and first-order sequence properties are still unavailable as a systematic open source library. Information-theoretic characterization that reveals higher-order temporal dependencies, e.g. entropy rate, excess entropy, auto-information structure, and Lempel–Ziv complexity, and, critically, their Markov-chain reference values that establish significance baselines, has only been formalized recently (von Wegner et al., 2017; von Wegner, 2026; von Wegner & Laufs, 2024). These methods are currently scattered across individual publications and an earlier Python package implementing a subset of these methods (von Wegner & Laufs, 2018b).

`mstsa` fills a gap and extends that work with a substantially expanded function set, C-accelerated implementations, and Numba JIT compilation. It is the only Python library that (i) implements the full set of information-theoretic complexity measures for symbolic sequences alongside their first-order Markov-chain predictions, enabling direct quantification of non-Markovian excess structure; (ii) provides the Kullback-information likelihood-ratio tests for Markovianity, stationarity, and transition-matrix symmetry (Kullback et al., 1962); (iii) includes different methods to estimate Hurst exponents from microstate sequences (von Wegner & Laufs, 2024); and (iv) models serial dependence between consecutive sojourn durations directly, via Autoregressive Conditional Duration models (Engle & Russell, 1998), rather than only characterizing each duration's marginal distribution in isolation. The ability to compare empirical measures to analytically and/or numerically derived Markov-chain reference values is a distinctive feature of the library (von Wegner, 2026).

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The library is standalone: it does not require MNE-Python and operates on any integer-valued symbolic sequence. This makes it applicable beyond EEG to any domain in which discrete time series arise, such as MEG-derived microstates, fMRI-defined sequences of network activations, and electrophysiological *in vitro* data. An earlier version (0.2.0) has been used in a series of peer-reviewed publications characterizing microstate dynamics during resting state, propofol anaesthesia, and sleep (Hermann & others, 2024; von Wegner et al., 2016, 2017; von Wegner, 2026; von Wegner & Laufs, 2018a, 2024).

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